Практикум A5

102 marks out of 144 are Paper 1, and the calculator share is 29%. There is nothing to compute; everything is solved by moving between three ways of writing the same number.

О практикуме

Доля 0.605 не подобрана на глаз: все 69 блоков темы разнесены между A5 и A6 поимённо, списки лежат в blocks карточек. A5 получил 42 блока и 144 балла, A6 — 27 блоков и 94 балла.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-a5-complex-forms

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Practicum A5: complex numbers, forms and arithmetic ℂ

The most «paper» topic of the archive after trigonometric equations: **102 marks out of 144 are Paper 1**, and the calculator share is 29%. There is nothing to compute; everything is solved by moving between three ways of writing the same number.

Material. 42 blocks and 144 marks from the AA HL archive, sessions May 2021 — November 2025. Every task below is a real past-paper question, cited.

What is here and what is in A6. Complex numbers make a large topic in the archive, 238 marks, and it splits in two by the substance of the technique rather than down the middle:

- **A5, this practicum** — the three forms, moving between them, arithmetic, the conjugate, modulus

and argument, conjugate roots of polynomials;

- **A6** — de Moivre's theorem, n-th roots, roots of unity, rotations and the geometry of the Argand plane.

The border runs where the arithmetic ends and raising to a power begins. If z^n shows up in the solution, it is A6.

How to work

1. Read the map of techniques. The first four are about the forms, the fifth is about the properties of a product, the sixth is about polynomials.
2. Work **on paper**. Draw the Argand plane: the quadrant the number lies in determines the argument, and by eye that is more reliable than an arctangent.
3. Enter the answer in whichever form suits you. The check compares **the number itself**, not the writing: `2*exp(I*pi/3)`, `2*(cos(pi/3) +`

`I*sin(pi/3))` and `1 + sqrt(3)*I` all count the same.

4. The last blocks are a recognition trainer and one question on a timer.

Difficulty marks: 🟢 the technique on its own · 🟡 the technique in a wrapper · 🔴 several techniques, or a whole exam question.

```
import sys
sys.path.append('..')          # from
practicum/number_algebra to practicum/kit/
import sympy as sp             # the escape
hatch: anything not in kit is in sp
from kit import *              # checks + I,
pi, sqrt, exp, re, im and the other names

language('en')                 # this
notebook is in English, and so are the checks

z, w = symbols('z w')
a, b = symbols('a b', real=True)

print('ready; sympy', sp.__version__)
print('the imaginary unit is I:', (1 + I)**2,
'   and pi, sqrt as usual:', 2*exp(I*pi/3))
```

Map of techniques

#	Technique	Trigger in the question	Gives
1	Arithmetic and the conjugate	a fraction with i in the denominator, «rationalise»	$a+bi$
2	Equate the parts	an equality of two complex expressions with unknowns	a system
3	Modulus and argument	«find $\lvert z\rvert$ and $\arg z$»	r, θ
4	Change of form	«in the form $re^{i\theta}$», «in Cartesian form»	another writing
5	Properties of a product	$\lvert z_1 z_2\rvert$, $\arg(z_1 z_2)$, the condition «$z \in \mathbb{R}$»	a relation
6	Conjugate roots	a polynomial with real coefficients	the other roots

Techniques 1–4 are one and the same skill seen from four sides: **moving freely between the three ways of writing a number.** The fifth and the sixth are built on top of it.

The three forms, between which everything happens:

$$z = a + bi \;=\; r(\cos\theta + i\sin\theta) \;=\; re^{i\theta}.$$

The rule of choice that saves half the work: **add in Cartesian, multiply in polar.** Addition and subtraction are trivial in $a+bi$ and meaningless in the polar form; multiplication, division and powers in the polar form come down to doing something to r and to θ separately.

Theory 1. The conjugate, and what it is for

The conjugate $\bar{z} = a - bi$ reflects the number in the real axis. Two of its properties do all the work:

$$z\bar{z} = a^2 + b^2 = |z|^2$$
$$\text{(always real)},$$
$$z + \bar{z} = 2a = 2\,\mathrm{Re}(z).$$

The first is the reason the conjugate is used at all. A complex number in a denominator is not an answer; to get rid of it, multiply numerator and denominator by the conjugate of the

denominator, and the denominator becomes real:

$$\begin{aligned}\frac{1}{2+3i} &= \frac{1}{2+3i}\cdot\frac{2-3i}{2-3i}\\ &= \frac{2-3i}{4+9}\\ &= \frac{2}{13}-\frac{3}{13}i.\end{aligned}$$

This is exactly the technique used to clear a root out of a denominator, and in the markscheme it is called rationalising.

Task 1 🟢 — into exponential form

Express $z_1 = 3-3i$ in the form $re^{i\theta}$, where $r>0$ and $-\pi<\theta\leq\pi$.

The order: the modulus by Pythagoras, then the quadrant from the signs, then the acute angle from the arctangent of the ratio.

Source: May 2025 TZ3, Paper 1, Q8(a) (3 marks, no calculator).

```
my_z1 = ...              # for example
5*exp(I*pi/3) or 5*(cos(pi/3) + I*sin(pi/3))

check_complex('Task 1', my_z1,
'dea41cd961fa')
```

Task 2 🟢 — one more change of form

Express $w = 1 + i\sqrt{3}$ in the form $re^{i\alpha}$, where $r > 0$ and $-\pi < \alpha \leq \pi$.

Source: November 2025 TZ3, Paper 1, Q11(a)(i) (2 marks, no calculator).

```
my_w = ...

check_complex('Task 2', my_w, '68abb3cc7597')
```

Theory 2. The argument and the quadrant

The modulus raises no questions: $|a + bi| = \sqrt{a^2 + b^2}$. The argument does, and almost always in the same way.

The formula $\arg z = \arctan \dfrac{b}{a}$ is **only valid in the right half-plane**, when $a > 0$. The

arctangent returns values in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, while an argument can be anything in $(-\pi, \pi]$. The numbers $1+i$ and $-1-i$ have the same ratio $b/a = 1$ but arguments $\frac{\pi}{4}$ and $-\frac{3\pi}{4}$.

A reliable order:

1. Mark the number on the plane and name the quadrant.
2. Find the **acute** angle $\alpha = \arctan\left|\frac{b}{a}\right|$.
3. Build it up to the right quadrant: I is α, II is $\pi - \alpha$, III is $\alpha - \pi$, IV is $-\alpha$.

The drawing here is not an illustration, it is the method itself. Checking takes a second: a number in the second quadrant must have its argument between $\frac{\pi}{2}$ and π.

Task 3 🟢 — modulus, argument, and deducing the form

Given $u = -1 + \sqrt{3}\,i$.

(a) Find $|u|$ and $\arg u$. **(b)** Hence show that $u = 2e^{2\pi i/3}$.

The number lies in the second quadrant — which is exactly where the arctangent formula gives the wrong answer.

Source: May 2023 TZ2, Paper 1, Q11(a) (3 marks, no calculator).

```
my_mod = ...
my_arg = ...              # the exact value

check_num('Task 3, |u|', my_mod, 6,
'd4735e3a265e')
check_expr('Task 3, arg u', my_arg,
'd99460aac5fb')
```

Task 4 🟢 — clearing the i out of the denominator

Given $w = \dfrac{4i}{1 + i\sqrt{3}}$.

Show that $w = \sqrt{3} + i$.

Source: November 2025 TZ1, Paper 1, Q11(a) (2 marks, no calculator).

```
my_w4 = ...               # the result in the
form a + b*I

check_complex('Task 4', my_w4,
'e6e0f8514921')
```

Task 5 🟢 — the modulus and argument of the result

For the same $w = \sqrt{3} + i$, find $|w|$ and $\arg w$.

Source: November 2025 TZ1, Paper 1, Q11(b) (3 marks, no calculator).

```
my_mod5 = ...
my_arg5 = ...            # the exact value

check_num('Task 5, |w|', my_mod5, 6,
'd4735e3a265e')
check_expr('Task 5, arg w', my_arg5,
'cb949ec26a0e')
```

Theory 3. Equating real and imaginary parts

The technique a third of the topic rests on. It comes from the definition of equality:

$$a + bi = c + di \quad \Longleftrightarrow \quad a = c \ \text{ and } \ b = d.$$

That is, **one complex equation is equivalent to two real ones**. Hence the recipe: if the unknowns in the question are real and the equation is complex,

split both sides into parts and get a system.

The order that saves time:

1. Write the unknown as $z = a + bi$ with a, b real.
2. Expand everything, remembering that $i^2 = -1$.
3. Collect separately everything without i and everything multiplying i.
4. Equate them to the corresponding parts of the right-hand side.
5. Solve the resulting system of two real equations.

Step 3 is where the marks go: a term like $qi \cdot i$ is real, not imaginary, and it is easy to leave it in the wrong pile.

Task 6 🟡 — a system out of one equation

Given that $z = 5 + qi$ satisfies the equation

$$z^2 + iz = -p + 25i,$$

where $p, q \in \mathbb{R}$, find the values of p and q.

Source: November 2023 TZ1, Paper 1, Q7 (5 marks, no calculator).

```
my_p = ...
my_q = ...

check_num('Task 6, p', my_p, 6,
'127bbd03a000')
check_num('Task 6, q', my_q, 6,
'd4735e3a265e')
```

Task 7 🟡 — an equation with a conjugate

Solve $z^2 = 2\bar{z}$ for $z \neq 0$. Find **all** the roots.

The same technique here, but with two surprises. First: after splitting into parts one of the equations factorises, and two cases appear. Second: there are three roots, although the equation looks quadratic.

Source: May 2023 TZ2, Paper 1, Q11(d) (5 marks, no calculator).

```
my_roots = [...]         # the list of all
roots, order does not matter

check_complex_set('Task 7', my_roots,
'25d575f62bcb')
```

Theory 4. Properties of a product and a quotient

In polar form multiplication splits into two independent actions:

$$|z_1 z_2| = |z_1| \cdot |z_2|,$$

$$\arg(z_1 z_2) = \arg z_1 + \arg z_2,$$

$$\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}, \qquad \arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2.$$

The moduli multiply, the arguments add. This is the only reason the polar form is needed at all.

How it is asked. Almost always through the condition «$z_1 z_2$ is real» or «is purely imaginary». It translates like this:

$$
\begin{aligned}
z \in \mathbb{R} &\iff \arg z \\
&= k\pi, \qquad z \text{ is purely imaginary} \\
&\iff \arg z \\
&= \frac{\pi}{2} + k\pi.
\end{aligned}
$$

That gives an equation in integers, and the smallest solution is asked for. Zero is an allowed value of the argument too — that is the one people forget.

Task 8 🟡 — the argument of a product

Given $z_1 = 1 + bi$ and $z_2 = (1 - b^2) - 2bi$, where $b \in \mathbb{R}$, $b \neq 0$.

(a) Find an expression for $z_1 z_2$ in terms of b. **(b)** Given that $\arg(z_1 z_2) = \dfrac{\pi}{4}$, find the value of b.

Part (a) is worth taking as far as «a positive number times something»: then in (b) the positive factor does not affect the argument, and everything comes down to one arctangent.

Source: May 2022 TZ1, Paper 1, Q9 (6 marks, no calculator).

```
my_b8 = ...              # the exact value

check_expr('Task 8', my_b8, 'e51873d4e350')
```

Task 9 🟡 — the condition for being real

Given complex numbers z and w with $|z| = 2, \quad \arg z = \frac{\pi}{5}, \quad |w| = 8, \quad \arg w = -\frac{2k\pi}{5}$, where $k \in \mathbb{Z}^+$.

Given that $zw \in \mathbb{R}$, find the smallest value of k and the corresponding value of zw.

Source: May 2021 TZ1, Paper 2, Q8 (7 marks, calculator).

```
my_k = ...
my_zw = ...

check_num('Task 9, k', my_k, 6,
'4e07408562be')
check_complex('Task 9, zw', my_zw,
'075ab20e88f6')
```

Task 10 🔴 — a square root through the parts

Given that $w = \sqrt{3} + i$ can be written as $w = (a + bi)^2$, where $a, b > 0$.

Find a^2 and b^2. The answers are **exact**.

Expand the square, equate the parts and get a system. It is non-linear, but it comes down to a quadratic in a^2. There is a short way too: $a^2 + b^2 = |a + bi|^2 = |w|$, and together with $a^2 - b^2 = \sqrt{3}$ that is already a linear system.

Source: November 2025 TZ1, Paper 1, Q11(d) (8 marks, no calculator).

```
my_a2 = ...              # the exact value
my_b2 = ...              # the exact value

check_expr('Task 10, a^2', my_a2,
'81b67c3de83c')
check_expr('Task 10, b^2', my_b2,
'0ac7146cef7e')
```

Theory 5. Conjugate roots of a polynomial

The fact the whole technique grows out of:

> If a polynomial has **real coefficients** and z_0 is a root of it, then $\bar{z}_0$ is a root as well.

The condition about real coefficients is essential, and in the question it is always written out explicitly. Without it the theorem is false.

The consequences that get used:

- complex roots come in **pairs**, so a polynomial of odd degree has at least one real root;
- knowing one complex root, you get a second one for free;
- a pair $z_0, \bar{z}_0$ gives a real quadratic factor $z^2 - 2\,\mathrm{Re}(z_0)z + |z_0|^2$.

After that Vieta comes in. For a polynomial $z^n + c_{n-1}z^{n-1} + \ldots + c_0$:

$$\sum z_i = -c_{n-1}, \qquad \prod z_i = (-1)^n c_0.$$

The sum of a conjugate pair is $2\,\mathrm{Re}(z_0)$, the product is $|z_0|^2$ — both real, and the unknowns come out of a single equation.

On a real question. A polynomial $z^4 + pz^3 + \ldots + 80$ with real coefficients has roots $3+i$, α, α^2 (with α real). The product of the roots is 80, so $(3+i)(3-i)\alpha\cdot\alpha^2 = 80$, that is $10\alpha^3 = 80$ and $\alpha = 2$. Then the sum of the roots: $-p = (3+i)+(3-i)+2+4 = 12$, hence $p = -12$. The imaginary parts cancelled by themselves — they always do.

Task 11 🟡 — the other roots of a cubic

Given that $u = -1+\sqrt{3}\,i$ is a root of the equation

$$z^3 + 5z^2 + 10z + 12 = 0,$$

find the other two roots.

Source: May 2023 TZ2, Paper 1, Q11(c)(i) (part of 9 marks, no calculator).

```
my_rest = [...]          # the two remaining
roots

check_complex_set('Task 11', my_rest,
'8f929b63c73d')
```

Trainer: name the technique

Nothing is solved here: name the technique with one code.

Code	Technique
`cart`	arithmetic, clear the i out of the denominator
`parts`	equate real and imaginary parts
`modarg`	find the modulus and the argument
`form`	change to another form
`prop`	properties of the modulus and argument of a product
`roots`	conjugate roots of a polynomial, Vieta

1. «Express $3-3i$ in the form $re^{i\theta}$»
2. «Write $2e^{i\pi/3}$ in Cartesian form»
3. «Find $|u|$ and $\arg u$ for $u=-1+\sqrt{3}i$»
4. «Show that $\dfrac{4i}{1+i\sqrt{3}}=\sqrt{3}+i$»

5. «Find $\arg w$, where $w = \sqrt{3} + i$»
6. «$z = 5 + qi$ satisfies $z^2 + iz = -p + 25i$. Find p and q»
7. «Solve $z^2 = 2\bar{z}$»
8. «Find $\arg(z_1 z_2)$ in terms of k»
9. «Given that zw is real, find the smallest k»
0. «$w = (a + bi)^2$, find a^2»
1. «$-1 + \sqrt{3}i$ is a root of a cubic. Find the others»
2. «Two roots of a quartic are $a + bi$ and $b + ai$. Find a»
3. «Simplify $\dfrac{1}{2 + 3i}$»
4. «Find p, given that the sum of the roots is 12»
5. «Show that $|z_1 z_2| = |z_1||z_2|$»

The target is 15 out of 15 in under two minutes.

```
answers = {
    1: '', 2: '', 3: '', 4: '', 5: '', 6: '',
7: '', 8: '',
    9: '', 10: '', 11: '', 12: '', 13: '',
14: '', 15: '',
}

KEY = {1: '07397d633f25', 2: '07397d633f25',
3: 'cbf396d2cbbb', 4: '1c42ddc0285b', 5:
'cbf396d2cbbb', 6: 'd887db09649d', 7:
'd887db09649d', 8: 'e835d0bf51a4', 9:
'e835d0bf51a4', 10: 'd887db09649d', 11:
'a4e3c20a3277', 12: 'a4e3c20a3277', 13:
'1c42ddc0285b', 14: 'a4e3c20a3277', 15:
'e835d0bf51a4'}
trigger_check(answers, KEY)
```

Task 12 🔴 — ON THE TIMER: 12 minutes

Close everything above. Paper 1 format, no calculator, 8 marks.

The equation $z^4 + 4z^3 + 8z^2 + 80z + 400 = 0$ has real coefficients. Given that $a + bi$ and $b + ai$ are two of its roots, where $a, b \in \mathbb{R}$, find the possible values of a.

A hint about the structure, not about the solution: the coefficients are real, so you

know not two roots but four. After that you need both of Vieta's formulas — one gives a linear relation, the other a quadratic one.

Source: May 2021 TZ1, Paper 1, Q7 (8 marks, no calculator).

```
# set a timer for 12 minutes
my_a_values = [...]      # all the possible values of a

check_complex_set('Task 12', my_a_values, '7de52fdac652')
```

Attempt log

Question	Answer
Did you finish inside 12 minutes?	
Did you write down all four roots straight away?	
Which of Vieta's formulas did you take first?	
Where did the time go?	

If you started expanding $(z-(a+bi))(z-(a-bi))\dots$ head-on, that works too and the markscheme accepts it, but it is

longer than writing down the sum and the product.

🔑 Solutions

Check not only the answer but that no step is missing: the markscheme pays for method.

Solution 1 — into exponential form

$z_1 = 3 - 3i$ lies in the **fourth** quadrant: the real part is positive, the imaginary part negative.

$$r = \sqrt{3^2 + (-3)^2} = \sqrt{18} = 3\sqrt{2}.$$

The acute angle: $\arctan\left|\frac{-3}{3}\right| = \arctan 1 = \frac{\pi}{4}$. In the fourth quadrant the argument is negative, so $\theta = -\frac{\pi}{4}$.

$$z_1 = 3\sqrt{2}\, e^{-i\pi/4}.$$

```
my_z1 = 3*sqrt(2)*exp(-I*pi/4)
```

Solution 2 — one more change of form

$w = 1 + i\sqrt{3}$ is in the **first** quadrant, so the arctangent gives the answer straight

away.

$$r = \sqrt{1+3} = 2, \qquad \alpha = \arctan\frac{\sqrt{3}}{1} = \frac{\pi}{3}.$$

$$w = 2e^{i\pi/3}.$$

```
my_w = 2*exp(I*pi/3)
```

Solution 3 — modulus, argument, and deducing the form

$u = -1 + \sqrt{3}i$ is in the **second** quadrant.

$$|u| = \sqrt{(-1)^2 + (\sqrt{3})^2} = \sqrt{4} = 2.$$

The acute angle is $\arctan\frac{\sqrt{3}}{1} = \frac{\pi}{3}$, and in the second quadrant the argument is built up as $\pi - \frac{\pi}{3} = \frac{2\pi}{3}$.

$$u = 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right) = 2e^{2\pi i/3}.$$

```
my_mod = 2
my_arg = 2*pi/3
```

The answer $-\frac{\pi}{3}$ is the standard mistake: that is exactly what a calculator gives for $\arctan(-\sqrt{3})$, because it does not know which quadrant the number is in.

Solution 4 — clearing the i out of the denominator

Multiply by the conjugate of the denominator:

$$\begin{aligned} w &= \frac{4i}{1 + i\sqrt{3}} \cdot \frac{1 - i\sqrt{3}}{1 - i\sqrt{3}} \\ &= \frac{4i(1 - i\sqrt{3})}{1 + 3} \\ &= \frac{4i - 4i^2\sqrt{3}}{4} \\ &= \frac{4\sqrt{3} + 4i}{4} \\ &= \sqrt{3} + i. \end{aligned}$$

```
my_w4 = sqrt(3) + I
```

The key step is $-4i^2\sqrt{3} = +4\sqrt{3}$: minus times minus gives plus, and the imaginary term becomes real.

Solution 5 — the modulus and argument of the result

$w = \sqrt{3} + i$ is in the first quadrant.

$$\begin{aligned} |w| &= \sqrt{3+1} \\ &= 2, \qquad \arg w \\ &= \arctan \frac{1}{\sqrt{3}} \\ &= \frac{\pi}{6}. \end{aligned}$$

```
my_mod5 = 2
my_arg5 = pi/6
```

Solution 6 — a system out of one equation

Substitute $z = 5 + qi$ and expand:

$$\begin{aligned} z^2 &= (5+qi)^2 \\ &= 25 + 10qi + q^2 i^2 \\ &= (25 - q^2) + 10qi, \end{aligned}$$

$$iz = i(5+qi) = 5i + qi^2 = -q + 5i.$$

Add them and collect the parts:

$$\begin{aligned} z^2 & \\ + iz \qquad\qquad &= \underbrace{(25 - q^2 - q)}_{\text{real}} \\ + \underbrace{(10q+5)}_{\text{imaginary}}\, i & \\ = -p + 25i. & \end{aligned}$$

Imaginary parts: $10q + 5 = 25 \Rightarrow q = 2$. Real parts: $25 - 4 - 2 = 19 = -p \Rightarrow p = -19$.

```
my_p = -19
my_q = 2
```

Note the order: the imaginary equation contains only q, so that is the one to start with. Starting with the real one you get an equation in two unknowns.

Solution 7 — an equation with a conjugate

Let $z = a + bi$. Then

$$z^2 = (a^2 - b^2) + 2abi, \qquad 2\bar{z} = 2a - 2bi.$$

Equate the parts:

$$a^2 - b^2 = 2a, \qquad 2ab = -2b.$$

The second equation factorises: $2b(a + 1) = 0$, so $b = 0$ or $a = -1$. Take both cases.

Case $b = 0$. Then $a^2 = 2a$, that is, $a = 0$ or $a = 2$. Zero is excluded by the question, so $z = 2$ remains.

Case $a = -1$. Then $1 - b^2 = -2$, so $b^2 = 3$ and $b = \pm\sqrt{3}$. That gives $z = -1 \pm \sqrt{3}i$.

$$z \in \{\, 2,\ -1 + \sqrt{3}i,\ -1 - \sqrt{3}i \,\}.$$

```
my_roots = [2, -1 + sqrt(3)*I, -1 -
sqrt(3)*I]
```

The temptation to divide the second equation by b costs two roots out of three here: you cannot divide by an expression that may be zero — exactly as in trigonometric equations.

A pleasant observation: the three numbers found are precisely the cube roots of 8. And so they should be: multiplying $z^2 = 2\bar{z}$ by z gives $z^3 = 2z\bar{z} = 2|z|^2$, and $|z| = 2$ for all three.

Solution 8 — the argument of a product

(a) Expand:

$$\begin{aligned} z_1 z_2 &= (1 + bi)\big((1 - b^2) - 2bi\big) \\ &= (1 - b^2) - 2bi + bi(1 - b^2) - 2b^2 i^2. \end{aligned}$$

Collect: the real part is $1 - b^2 + 2b^2 = 1 + b^2$, the imaginary part $-2b + b - b^3 = -b(1 + b^2)$. So

$$z_1 z_2 = (1 + b^2)(1 - bi).$$

(b) The factor $1 + b^2$ is **strictly positive**, so it does not affect the argument:

$$\arg(z_1 z_2) = \arg(1 - bi) = \frac{\pi}{4}.$$

The number $1 - bi$ has real part 1 and imaginary part $-b$. An argument of $\frac{\pi}{4}$ means the parts are equal and both positive: $-b = 1$, that is, $b = -1$.

```
my_b8 = -1
```

Check: at $b = -1$ we get $z_1 z_2 = 2(1 + i) = 2 + 2i$, and $\arg(2 + 2i) = \frac{\pi}{4}$.

Solution 9 — the condition for being real

The modulus: $|zw| = |z||w| = 2 \cdot 8 = 16$.

The argument: $\arg(zw) = \arg z + \arg w = \frac{\pi}{5} - \frac{2k\pi}{5} = \frac{(1 - 2k)\pi}{5}$.

A number is real exactly when its argument is a multiple of π:

$$\frac{(1-2k)\pi}{5} = n\pi \implies 1-2k = 5n.$$

Go through the n: at $n=-1$ we get $1-2k=-5$, that is, $k=3$ — a positive integer. At $n=1$ we get $k=-2$, at $n=0$ a fraction. The smallest value that works is $k=3$.

Then $\arg(zw) = \dfrac{(1-6)\pi}{5} = -\pi$, and

$$zw = 16(\cos(-\pi) + i\sin(-\pi)) = -16.$$

```
my_k = 3
my_zw = -16
```

Solution 10 — a square root through the parts

$$(a+bi)^2 = a^2 - b^2 + 2abi = \sqrt{3} + i.$$

Equate the parts: $a^2 - b^2 = \sqrt{3}$ and $2ab = 1$.

The short way. Take the modulus of both sides of the original equality: $|a+bi|^2 = |w|$, that is, $a^2+b^2=2$. Together

with $a^2 - b^2 = \sqrt{3}$ this is a linear system in a^2 and b^2:

$$a^2 = \frac{2 + \sqrt{3}}{2}, \qquad b^2 = \frac{2 - \sqrt{3}}{2}.$$

The direct way. From $2ab = 1$ we get $b = \frac{1}{2a}$, and substituting gives $a^2 - \frac{1}{4a^2} = \sqrt{3}$, that is, $4a^4 - 4\sqrt{3}\,a^2 - 1 = 0$. As a quadratic in a^2:

$$\begin{aligned} a^2 &= \frac{4\sqrt{3} \pm \sqrt{48 + 16}}{8} \\ &= \frac{4\sqrt{3} \pm 8}{8} \\ &= \frac{\sqrt{3} \pm 2}{2}. \end{aligned}$$

The condition $a > 0$ leaves the plus sign, and both ways agree.

```
my_a2 = (2 + sqrt(3))/2
my_b2 = (2 - sqrt(3))/2
```

Check: $a^2 - b^2 = \sqrt{3}$ ✓, and $2ab = 2\sqrt{a^2 b^2} = 2\sqrt{\frac{4-3}{4}} = 1$ ✓.

Solution 11 — the other roots of a cubic

The coefficients are real, so the conjugate $-1-\sqrt{3}i$ is a root as well. That is the first answer and the first mark.

The third root c is real (in a cubic the complex roots come as a pair). From the sum of the roots:

$$\begin{aligned}&(-1+\sqrt{3}i)\\&\quad+(-1-\sqrt{3}i)\\&\quad+c\qquad\qquad=\\&\quad-5\\\implies&\ -2+c\\=&\ -5\\\implies&\ c\\=&\ -3.\end{aligned}$$

Or from the product: $(-1+\sqrt{3}i)(-1-\sqrt{3}i)\,c=-12$, where the first factor is $1+3=4$, so $4c=-12$ and $c=-3$.

$$z\in\{\,-1+\sqrt{3}i,\ -1-\sqrt{3}i,\ -3\,\}.$$

```
my_rest = [-1 - sqrt(3)*I, -3]
```

Solution 12 — the task on the timer

The coefficients are real and two roots are given: $a+bi$ and $b+ai$. So there are two more — their conjugates: $a-bi$ and $b-ai$.

The sum of the roots equals minus the coefficient of z^3, that is, -4:

$$\begin{aligned}
&(a+bi)\\
&\quad+(a-bi)\\
&\quad+(b+ai)\\
&\quad+(b-ai) \quad = 2a\\
&\quad+2b\\
&=-4\\
&\implies a+b\\
&=-2.
\end{aligned}$$

The product of the roots equals the constant term (the degree is even), that is, 400. The conjugate pairs give squared moduli:

$$\begin{gathered}(a+bi)(a-bi)\\(b+ai)(b-ai)=(a^2+b^2)(b^2+a^2)=(a^2+b^2)^2=400,\end{gathered}$$

hence $a^2 + b^2 = 20$ (a modulus is non-negative, so only the plus sign).

Solve the system. From the first, $b = -2 - a$; substituting:

$$\begin{aligned} a^2 + (a+2)^2 &= 20 \\ &\Longrightarrow\ 2a^2 + 4a - 16 \\ &= 0 \\ &\Longrightarrow\ a^2 + 2a - 8 \\ &= 0, \end{aligned}$$

$$a = 2 \quad \text{or} \quad a = -4.$$

```
my_a_values = [2, -4]
```

Check: at $a = 2$ we get $b = -4$, and $a^2 + b^2 = 4 + 16 = 20$ ✓. The pair $(a, b) = (-4, 2)$ gives the same four roots in a different order, so both values of a really are possible.